

*I will be putting my test scan files in a folder called Blender testing in our repo

Space

This scan mode utilizes iPhone LiDAR scanner to create a floorplan and mesh.



- I like how it gives a visualization of where it's scanning the mesh/polygons like in the image
- Better at scanning and understanding the layout/floorplan of buildings but the overall scan quality is only a little better than Scaniverse.
 - Many holes in the scan where there shouldn't be and objects that aren't boxy or too small don't have a very good depiction/rendering in the scan.
 - Does well at scanning the overall textures and appearance and putting them on the scanned objects
- Needs lots of patience and slow panning to get the scan but the inconsistencies and blemishes in the scan felt pretty unavoidable for me (at least for the area I was trying to scan)
 - Hard to try and balance correct amount of scanning because some areas that didn't get enough scanning looked worse but sometimes too much scanning caused weird overlaps and disconnects in certain spots.
- The main weakness of this scan is reflective or metallic surfaces (e.g. windows, mirrors, shiny metal). I'm not sure how well this would turn out when we scan windows on buildings.
- Overall: Good at mapping out general building structure but falls short in terms of details and quality. The holes and inconsistencies may be able to over come with clean up in blender but that works feels like it may be time consuming to get a good product.

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Floorplan



- This mode creates a scan of a much more simplified version of the rooms you are scanning.
- Because a detailed mesh isn't being made like in Space scans there are no hole or mesh inconsistencies
- Only main issue I can see for this option is texture. This scan gives a blank slate for what the walls and objects in the scan should look like instead of getting the actual appearance from the real life objects like in the Space scan. Not sure if we want to try creating our own art are making textures out of photos of the objects it is trying to depict.
 - Since this is a floorplan, if we are trying to scan outside I'm not sure how well it will capture a sloped floor (e.g. the road from security up to feldman horn) if we want an area like that for our map
- The objects in my scan were pretty close to how they looked in real life but it seems like the scan might just choose the closest object in its catalog of prefabs and place it where the scanned object is. This wasn't a huge concern for me but wanted to note that this may be a source of very minor inconsistencies.
- Huge positive is that if you import this as a .obj file into Blender, it allows you to move around individual objects like a desk which isn't possible for Space scan.

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Use:

- The scanning is pretty self explanatory and the app gives good explanations, directions and tutorials.
- Good overview of the whole app: https://www.youtube.com/watch?v=Ibcl_vGJj_I
- Playlist of a bunch of tutorial videos if you want but I don't think they will really be necessary:
<https://www.youtube.com/watch?v=gZ6AWrzIx6c&list=PLqnRz-4Awhm7MXTpkgq9paJLC3ONJEBUX>